

Learning rate, regularization, and tornado EF ratings

Jake Van Slyke
University of Oklahoma

Data and experiment design

18,639 rated tornadoes, 2010–2025. Six classes: EF0–EF5.

Training	Validation	Test
2010–2019	2020–2022	2023–2025
11,771 events	3,117 events	3,751 events

Three hidden layers: **128, 64, 32**. Same seed and 50 epochs.

Compare **SGD learning rates, L2, and dropout**.

EF describes surveyed damage. Inputs include radar, warnings, final track, and land cover.

SGD at 0.01 gave the best macro F1

Run	Validation accuracy	Validation macro F1
Adam, 0.001	0.577	0.445
SGD, 0.1	0.439	0.324
SGD, 0.01	0.555	0.448
SGD, 0.001	0.532	0.375

0.1 was erratic. At 0.001, learning was still slow after 50 epochs.

Final-epoch validation results. Macro F1 covers the five classes present in validation.

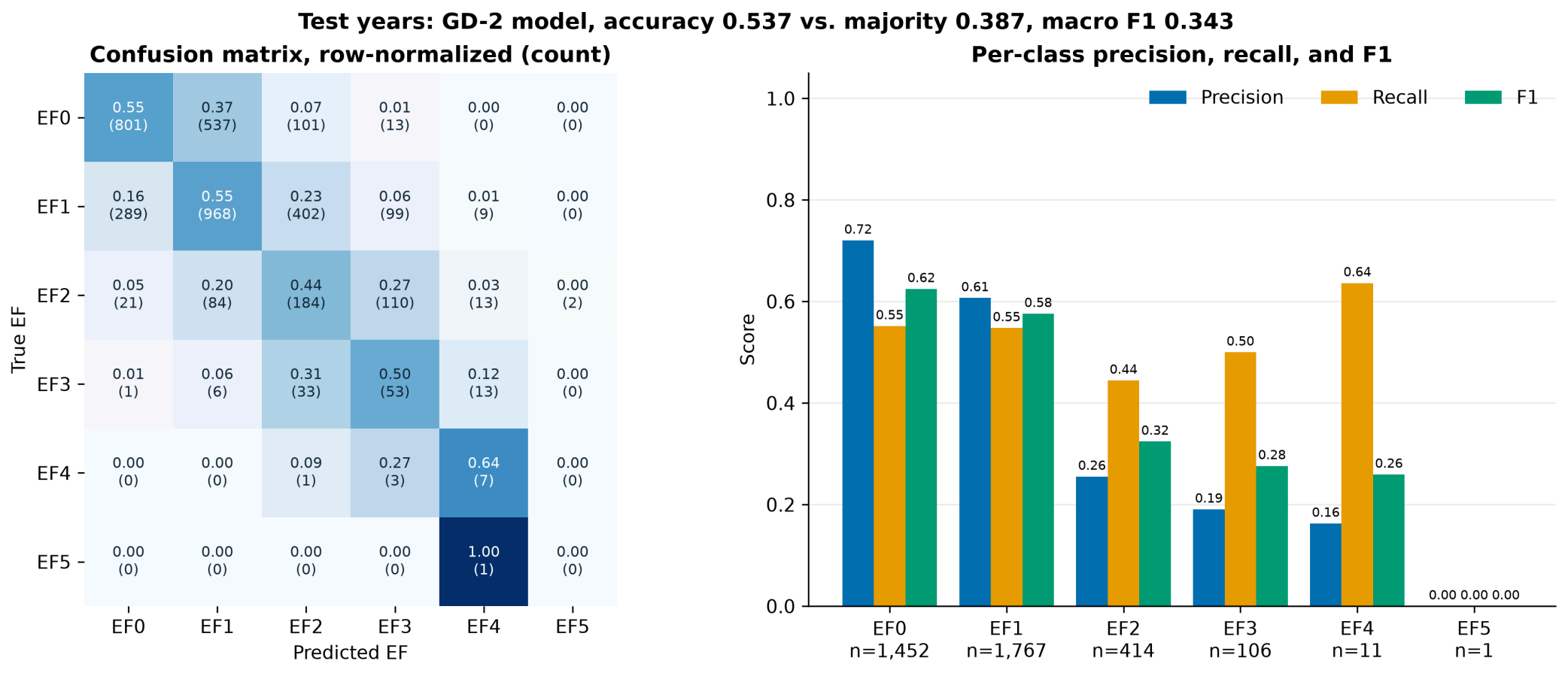
My “aha”: a smaller gap did not mean better F1

Adam run	Accuracy gap ¹	Validation loss	Macro F1
Baseline	19.6 pp	2.53	0.445
L2, 0.001	15.6 pp	1.59	0.428
Dropout, 0.2	11.5 pp	1.18	0.436
Dropout, 0.5	9.5 pp	0.88	0.430

Dropout 0.5 reduced the gap, but lowered validation accuracy.

¹Training minus validation accuracy, evaluated with dropout off. Loss includes class weights and any L2 penalty.

Test results: low precision for severe ratings



SGD 0.01 on 3,751 later events. Test loss: 1.072. Six-class macro F1: 0.343.

Final-track information contributed most

Features removed	Validation macro F1
None	0.448
Radar	0.400
Final track	0.289
Land cover	0.413
Final track and land cover	0.284

Track dimensions share information with the damage-based rating.

Same split and training settings. Validation comparisons only.

What remains unclear

Can patterns across EF ratings help us learn from rare severe events?

- Add **richer data** on storm conditions and exposure.
- Revisit **ordinal models** and other methods for class imbalance.
- Compare across seeds, then evaluate on independent later events.

EF5 support is only 7/0/1 across train/validation/test. Earlier ordinal and ensemble trials have not established a robust gain.